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## A Green Energy Company

### Technology Overview

January 12, 2011



## About Us

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Nanodify® provides the breakthrough synthesis of nano catalysts for hydrogen production, fuel cells, commodity chemicals, and biofuel, from renewable feedstocks.

### **Our focuses:**

- Nano-Nickel catalyst for hydrogen production and fuel cells
- Titanium Photocatalyst for biomass reforming and hydrogen production
- Fuel cell technology for electric vehicle and consumer power generation

**Two patents filed and additional patents ready to be filed**



# The Company

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Nanodify® is a Delaware Corporation, established in May, 2010 through the merge of CPNano, Inc. and Nanodify, Inc., founded in 2005.

## Summary of Nanodify:

1. We eliminate precious metals and high temperature in catalysts (huge cost savings)
2. We have very high yield photosensitive hydrogen catalysts (unique and patent pending)
3. Our photocatalysts can make hydrogen for power generation and
4. Can generate hydrogen aboard cars to be used with our fuel cells for clean cheap fuel.

**Nanodify technology is highly disruptive to hydrogen producing industry.**



# Product & Technology

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Nanodify® provides five different products/technologies:

- 1) Titanium Photocatalyst for commodity chemicals,
- 2) Titanium Photocatalyst to produce biofuel from renewable feedstock,
- 3) Nano-Nickel catalyst for hydrogen production in fuel cells for electric vehicles,
- 4) Nano-Nickel catalyst for hydrogen production used in consumer power generation
- 5) Fuel cell module for electric vehicle and consumer power generation.

[Nanodify Technology Demo](#)



# Proprietary Nano-Nickel Catalyst Technology

- **Substantial value driver** in the development of fuel cells and hydrogen production
- Custom-tailored nano-nickel catalysts for the specific needs of our customers' catalytic selectivity applications
- Particle size in the range of 20-50 nanometers with a surface area having a range of 40– 60m<sup>2</sup>/gm
- Excellent electrical conductance material in support of hydrogen production and fuel cells.



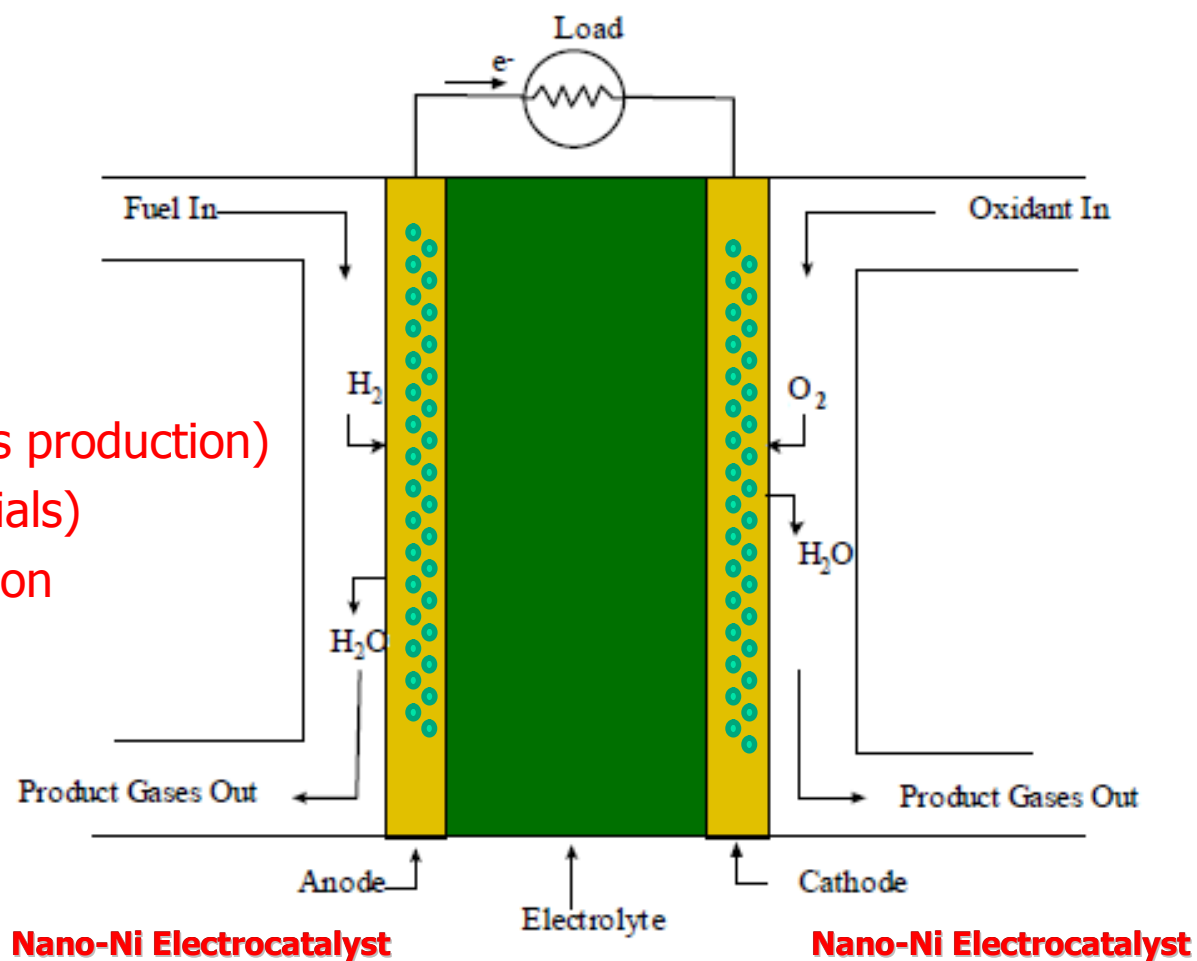
Photo shows the efficiency of the Nano-Nickel Catalyst. Even the low level flashlight shown above produces an abundance of hydrogen gas.

**Patent to be filed by Morgan, Lewis & Bockius LLP**



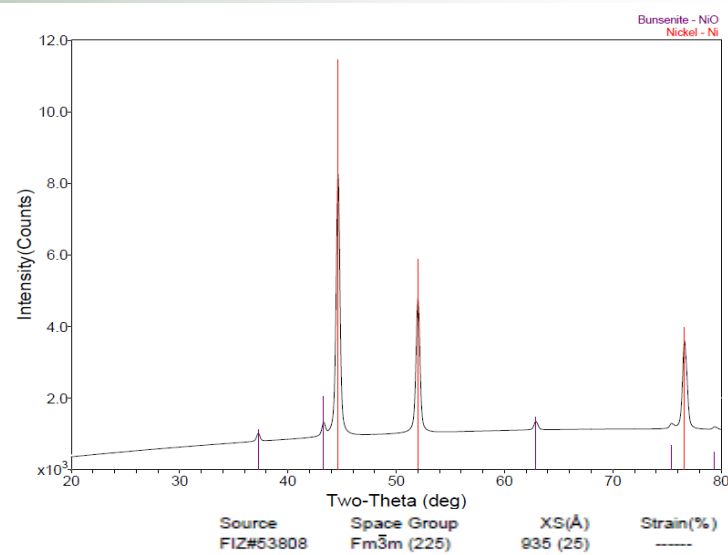
# Fuel Cell Electrolyzer

- Highly scalable (mass production)
- Low-cost (raw materials)
- Mass commercialization

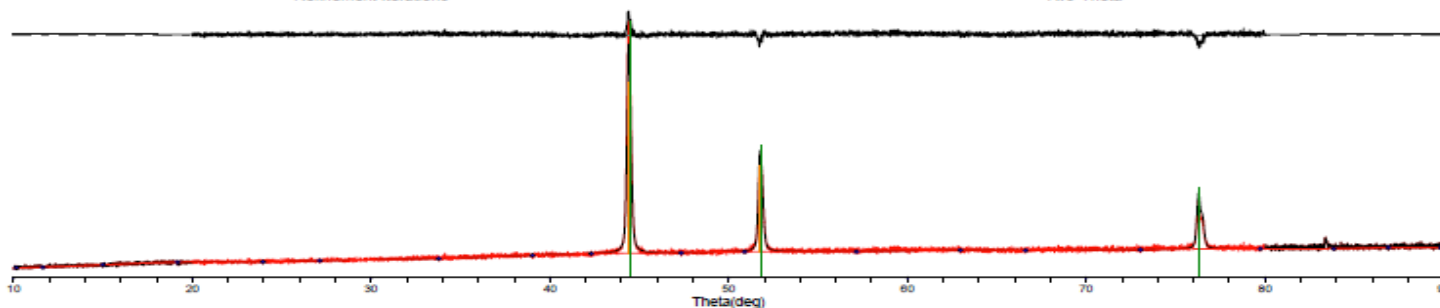
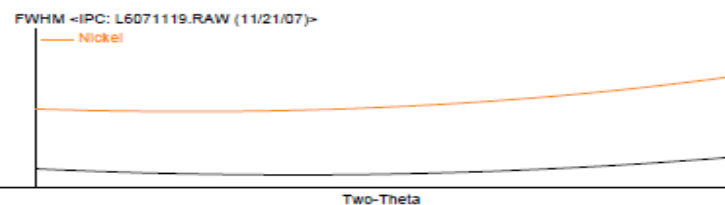
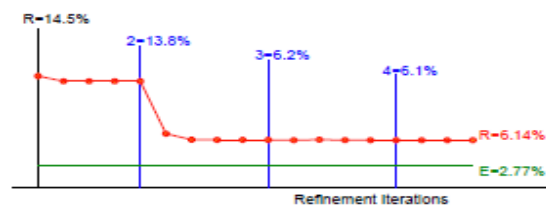




# Nano Nickel Catalyst



Phase ID (1)  
 Nickel - Ni  
 NOTE: Fitting Halted at Iteration 16(4): R=6.12% (E=2.77%, R/E=2.21, P=6, EPS=0.5)





# Titanium Photocatalysts Technology

- Custom-tailored heterogeneous Titanium photocatalyst for biomass reforming and sugar derived chemicals
- Low-cost catalyst for used in the production of hydrogen, commodity chemicals, and biofuel
- Renewable alternative solution and brings the world one vital step closer to reducing our fossil fuel consumption

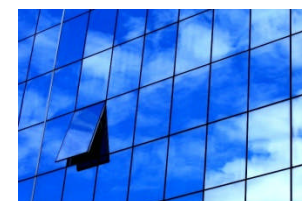
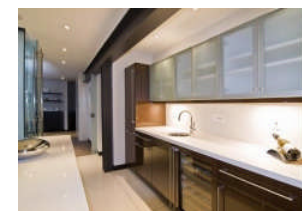


**Synthetic Bio-Ethanol Process  
Main Feedstock "Sugars"**



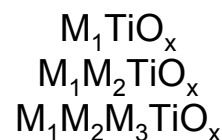
**Bio-Ethanol**

**Others Market  
Opportunity  
(cleaning, disinfecting,  
and coating)**



**Patent filed by Morgan, Lewis & Bockius LLP**

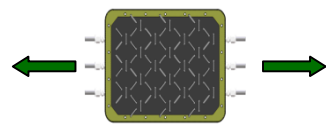






# Market Opportunity

- 1) Commodity chemicals, biofuel, and hydrogen production
- 2) Fuel cells for electric vehicle
- 3) Consumer power generation



- Hydrogen/fuel-cells are a **multi-trillion dollars industry**
- Market size for nano materials and nano-enabled products can go as high as **\$3 trillion dollars by 2015** (Lux, 2009).
- Current market size for this sector is only **2% of the \$3 trillion dollars.**
- Congress (2009) passed more than **\$90 billions dollars** legislation for clean tech development.



# Competitors



- CRI Catalyst,
- Matrostech,
- Sued Chemie Catalyst,
- Umicore,
- Inco,
- Haldor Topsoe,
- Nextech Materials,
- Headwaters Technology,
- Integran Technology,
- N.E. Chemcat ...

## No competitor provides

Very high throughput

Photo activation

Low cost / low temperature materials

**Nanodify patent pending products provide all of these key features.**



# Nanodify Competitive Advantage

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- **Highly scalable** to support mass production
- **Low-cost** (raw materials, manufacturing process)
- **Significantly lower cost** than most existing solutions such as platinum, gold, etc
- Easily be converted to **mass commercialization**
- **Minimal technical risk** to mass commercialization



# Business Development

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- Technology development and manufacturing custom tailored heterogeneous catalysts for hydrogen production, fuel cells, biomass reforming and sugar derived chemicals
- Technology development for fuel cells in electric vehicle
- Technology development for sugar derived chemicals
- Technology development biomass reforming
- Joint venture in co-developing the next generation fuel cells for electric vehicle and consumer power generation



# Operations

- Nanodify's research and development headquarter is located at Tully Business Center, 1340 Tully Road, Suite 308, San Jose, California, 95122. The Tully Business Center is a good suit for conducting nano materials research and development, hydrogen production, and fuel cell. The Tully Business Center was designed as a high tech facility with a modular research laboratory, chemical permits, on-site safety, facility support, and security. Nanodify's leases approximately 1728 sq. ft. of space at this location.



1340 Tully Road, Suite 308, San Jose, CA

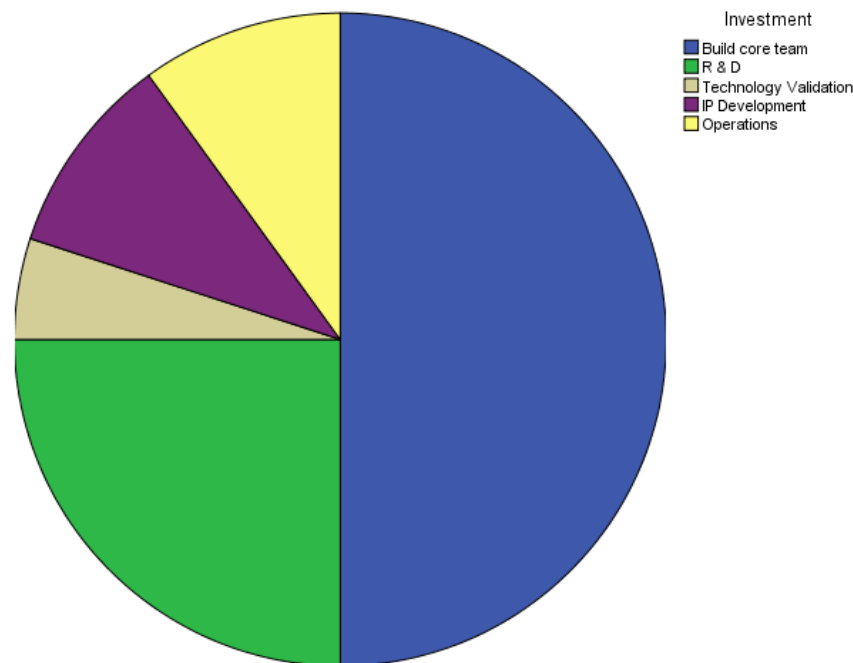
**Move into a production scale facility by 2012**





## Use of Fund

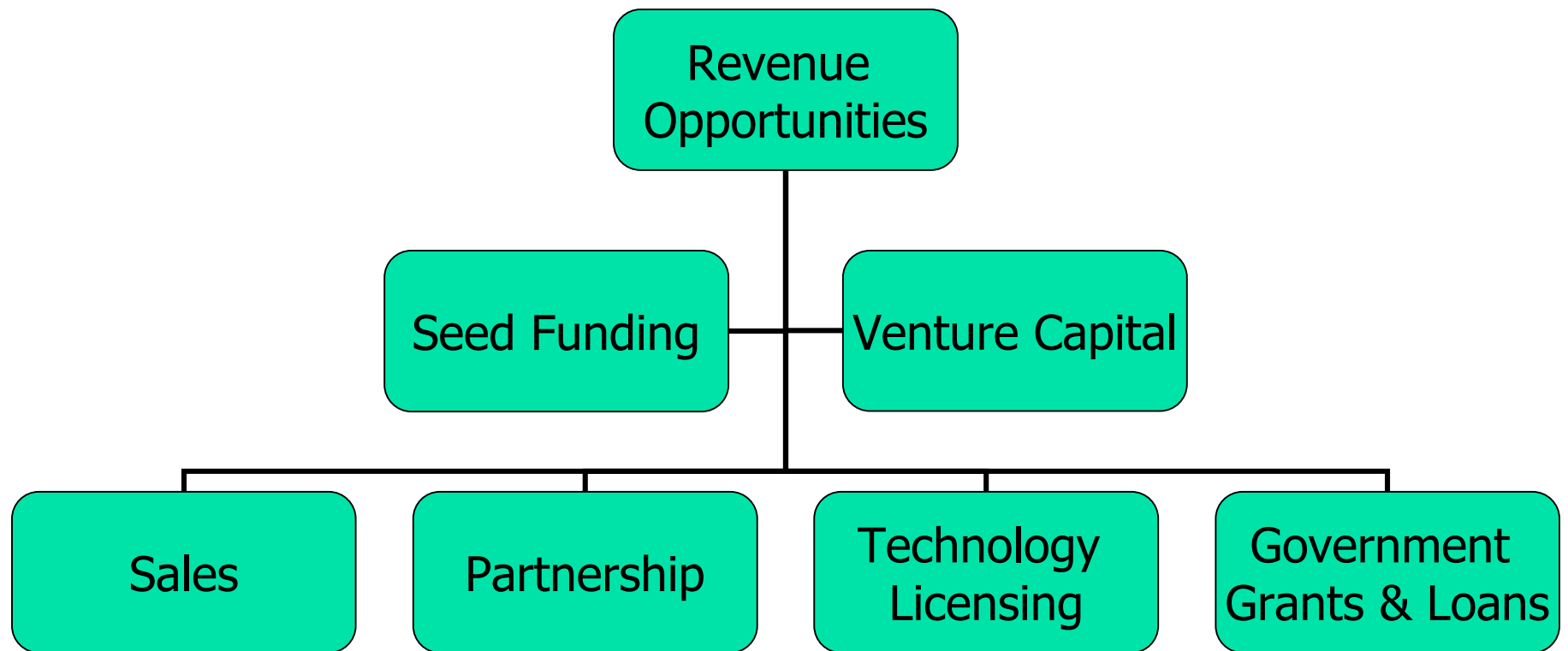
- Technology development and manufacturing custom tailored heterogeneous catalyst for hydrogen production, fuel cells, biomass reforming and sugar derived chemicals
- Technology development for fuel cells in electric vehicle and consumer power generation
- Technology development for sugar derived chemicals
- Technology development biomass as reforming
- Intellectual Property (IP) development



**Seeking \$5 million dollars**



# Revenue Opportunities







## Go to Market Strategy

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- Generate income form the direct sale of Nanodify's products
- Form strategic partnerships, and
- Licensing the technology to key end users.



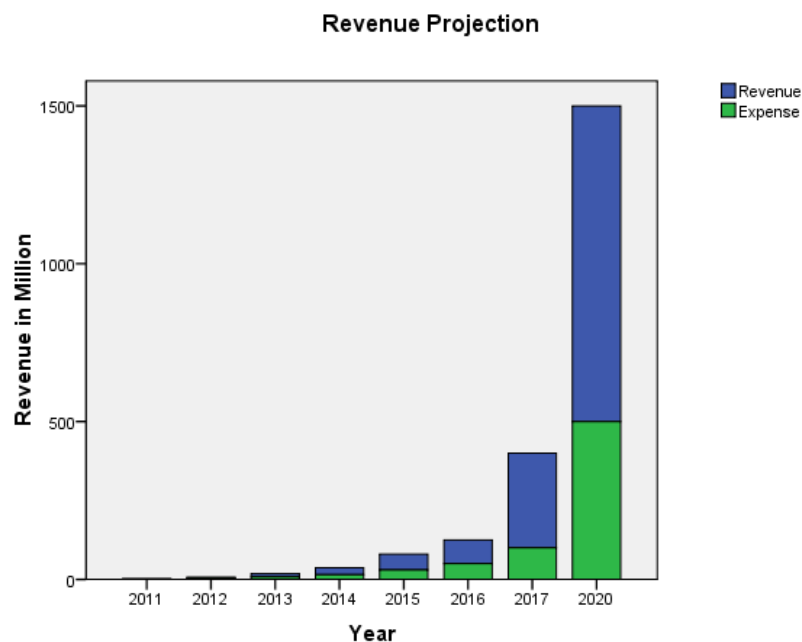
## ROI and Exit Strategy

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- Acquisition or IPO
- Tremendous Potential Return on Investment
- The future looks bright for a company with a focus on green technology.



# Revenue Projection



**\$50 million dollars  
surplus by 2014**

The product sales, strategic partnerships and licensing:

- Nano-Nickel catalyst, Titanium Photocatalyst, and custom tailored nano catalysts
- The development and production of fuel cell modules for electric vehicles

Potential Customers:

DOE, Petroleum Companies, Hydrogenics, Intelligent Energy, Horizon Fuel Cell, Nuvera, UltraCell Power, Air Product, Air Liquide, Praxair, Dupont Fuel Cell, Global Hydrogen, ITM Power, H2gen, Panasonic, Toyota, GM, etc

**Revenue surplus and self-sustain by 2014**



# Why Investment In Nanodify

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- Nanodify revolutionary catalyst materials are **low-cost** for hydrogen production, fuel cells, and electrolysis.
- Nanodify's disruptive Nano-Nickel catalyst technology is a **substantial value driver** for fuel cells than high-cost platinum, gold, etc.
- Nanodify's fuel cell module can be scaled up for **consumer power generation**.
- Nanodify's Nano-Nickel catalyst materials provides **high efficiency, low cost, and low environmental impact** for hydrogen production and fuel cells.
- Nanodify's Titanium Photocatalyst materials is fully support **biomass reforming** to hydrogen gas.
- **Tremendous Return on Investment**



# Company Representative

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## **Executive Management Team:**

- Dr. Jim Pham, Ph.D., CEO
- Mr. Vincent Pham, CTO & Founder, Scientist
- Dr. Ang-Ling Chu, Ph.D., Vice President of Operations & Scientist
- Mrs. Van-Hanh Nguyen, MBA, J.D., CFO
- Mr. John Zajac, Vice President of Engineering
- Mr. Tom Huynh, Vice President of Production
- Mr. Nhut Tran, Director of Business Development
- Dr. Shao-Sheng Dong, Ph.D., Polymer Scientist
- TBD, Sales and Marketing

## **Other Key Nanodify Representative:**

- Patent Counsel: Dr. Tuan Nguyen, Ph.D., J.D., Morgan, Lewis & Bockius LLP
- Corporate Attorney: John Park, Morgan, Lewis & Bockius LLP
- Analytical Chemistry Service: Dr. J. P. Hsu, Ph.D., Smart Chemistry Corporation
- The Blocc Group, LLC: Kevin Hoang, COO, The Asia Market

## **Board of Advisors:**

- Dr. Joseph Adwere-Boamah, Ph.D., Academic Scholar, UC Berkeley
- Dr. Jason Huang, Ph.D., Mechanical Engineering, Queen Mary, University of London
- Dr. Nobuyuki Kambe, Ph.D., MIT; co-Founder of Nanogram Devices, Neophontics, and Nanogram Corporation.
- Dr. Tien D. Pham, Ph.D., EE, University of Toronto; co-Founder Radilan
- Dr. Edward Shenk, Ed.D., Academic Scholar, University of Oregon
- Dr. George Stamos, Ed.D., Academic Scholar, Northern Arizona University
- Mr. Daniel Barr, MBA, Director of Transportation Systems for the State of Ohio
- Mr. George Reis, Engineering Executive, BSEE, Iowa State University
- Mr. Hung Le, Technology Veteran and Executive; MSEE and BSEE from MIT with specialization in device physics



# Thank You!

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## Q & A